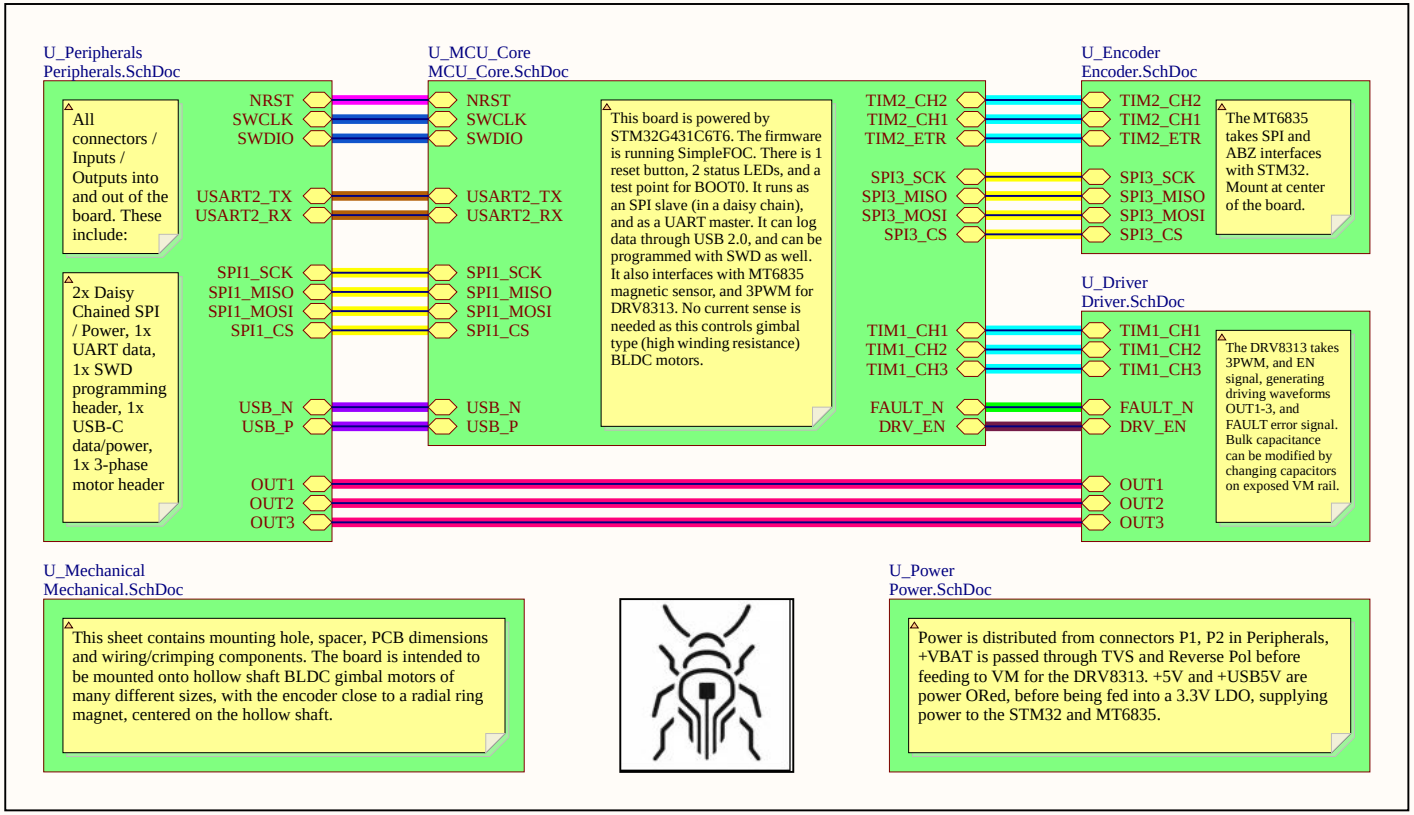
UQ Biorobot New Tracker
Gimbal Driver Module

System Overview - All Sheets

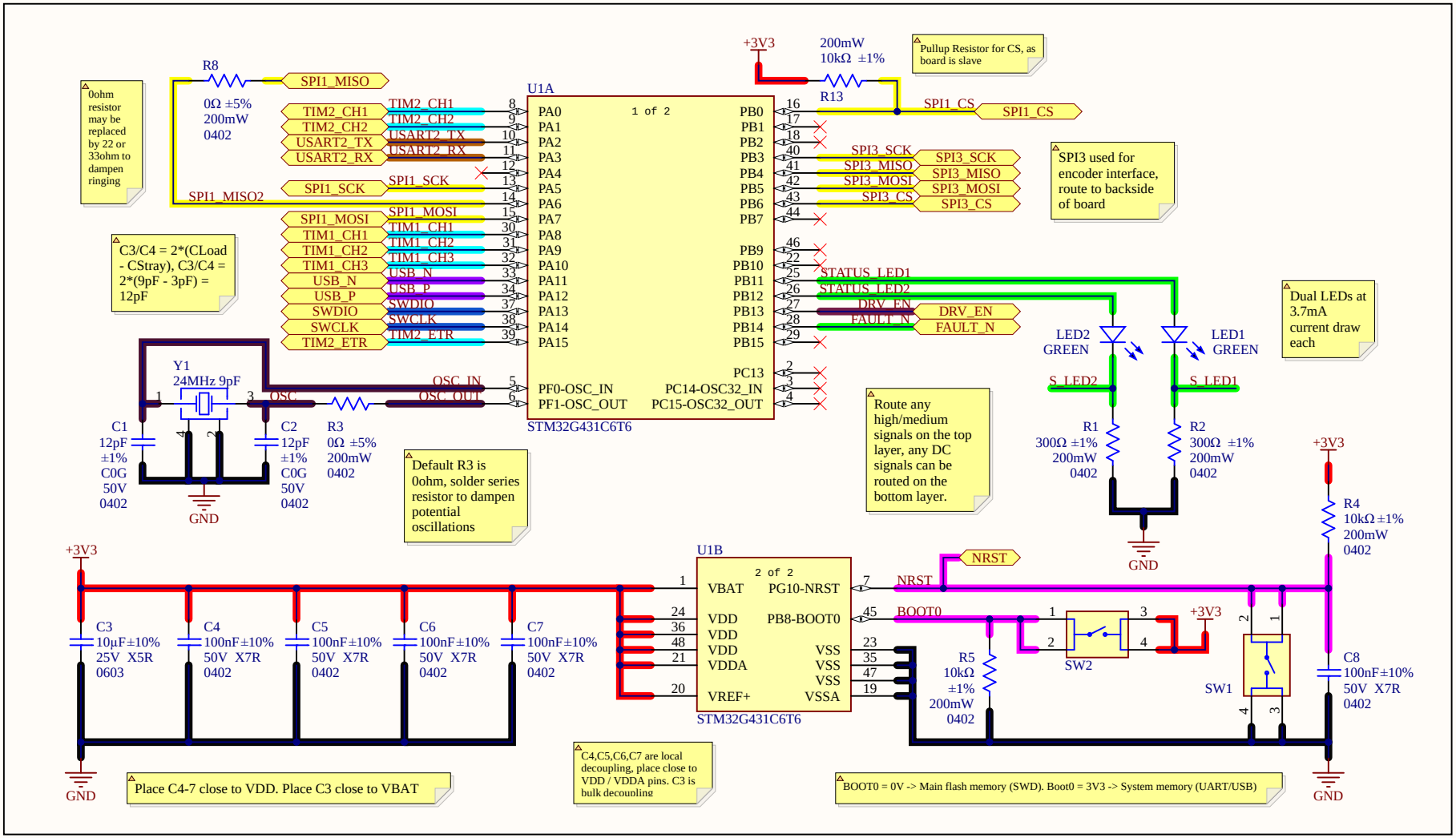


Stackup / Impedance Profiles

Cannot open file C:\Users\acll2\Desktop\NewTracker\NewTracker\Stackup_Impedance.png. File does not exist.

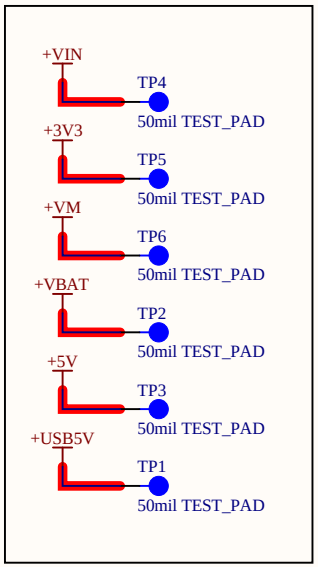
Title: Overview		
Size: A4	Number: 1	Revision: 1
Date: 10/07/2026		Sheet 1 of 7
File: C:\Users\..\Overview.SchDoc		Drawn By: Xuyang Liu

STM32G431C6T6 Microcontroller

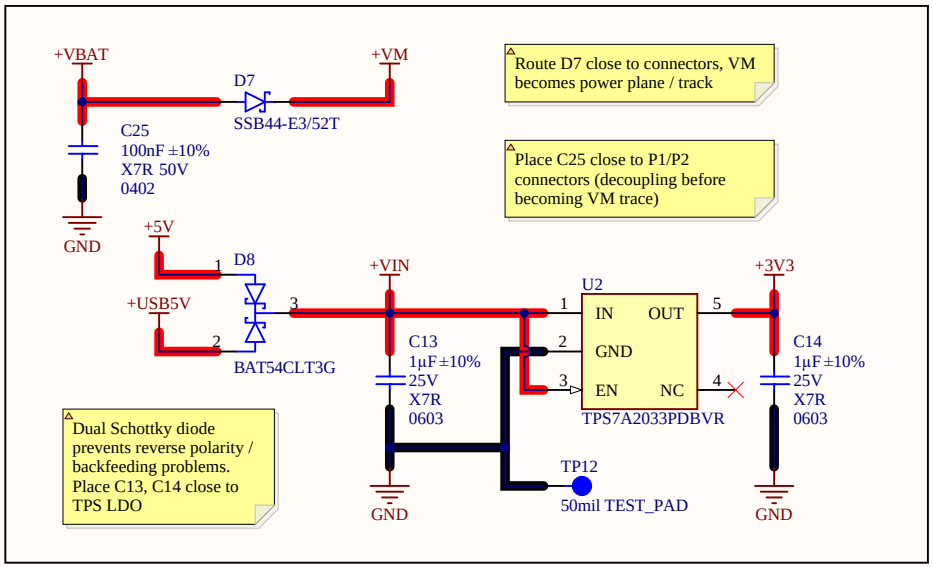


Title: MCU_Core		
Size: A4	Number: 2	Revision: 1
Date: 10/07/2026		Sheet 2 of 7
File: C:\Users\..\MCU_Core.SchDoc		Drawn By: Xuyang Liu

Voltage Rail Test Points

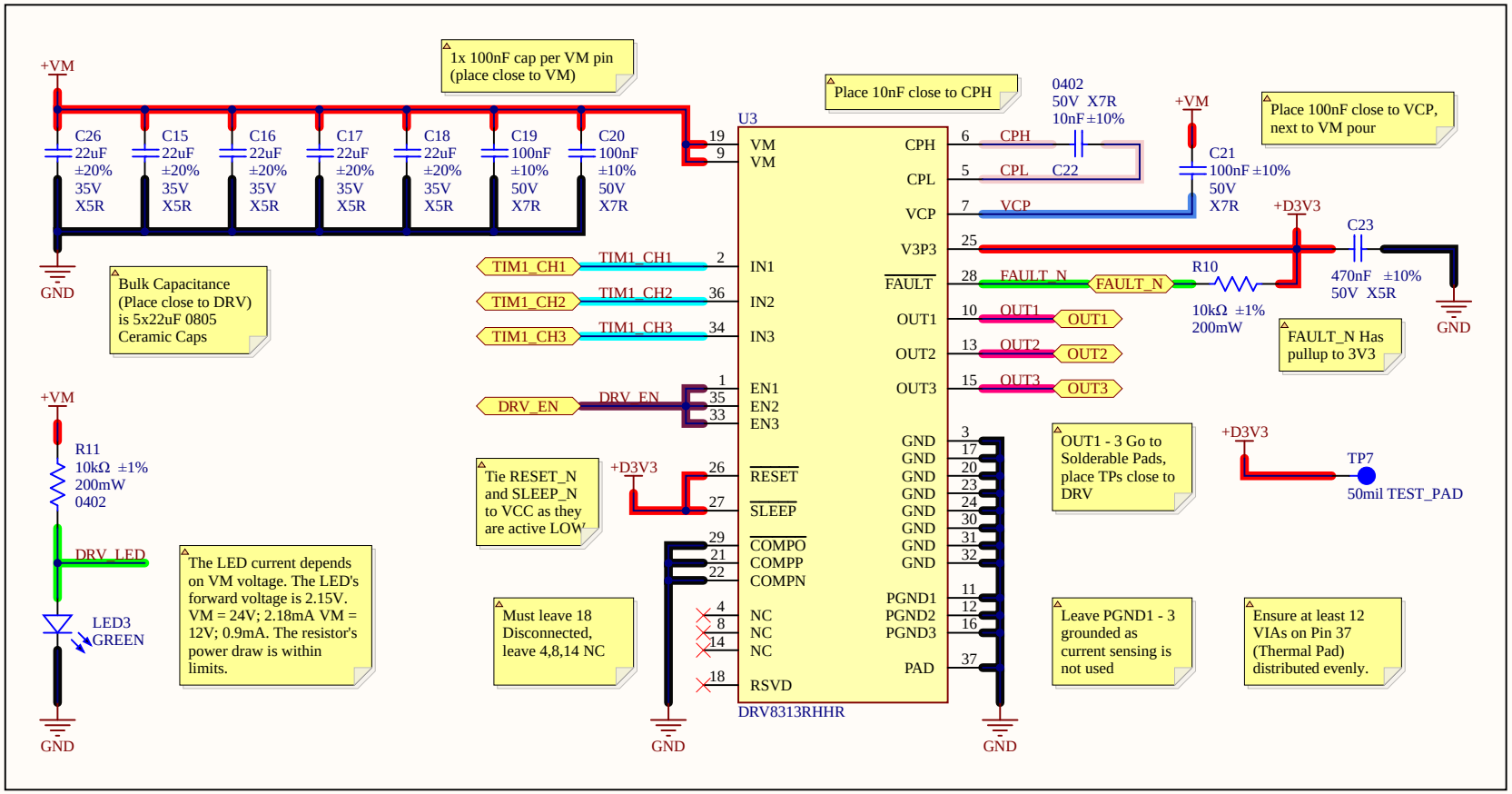


5V Power ORing and Reverse Polarity Protection



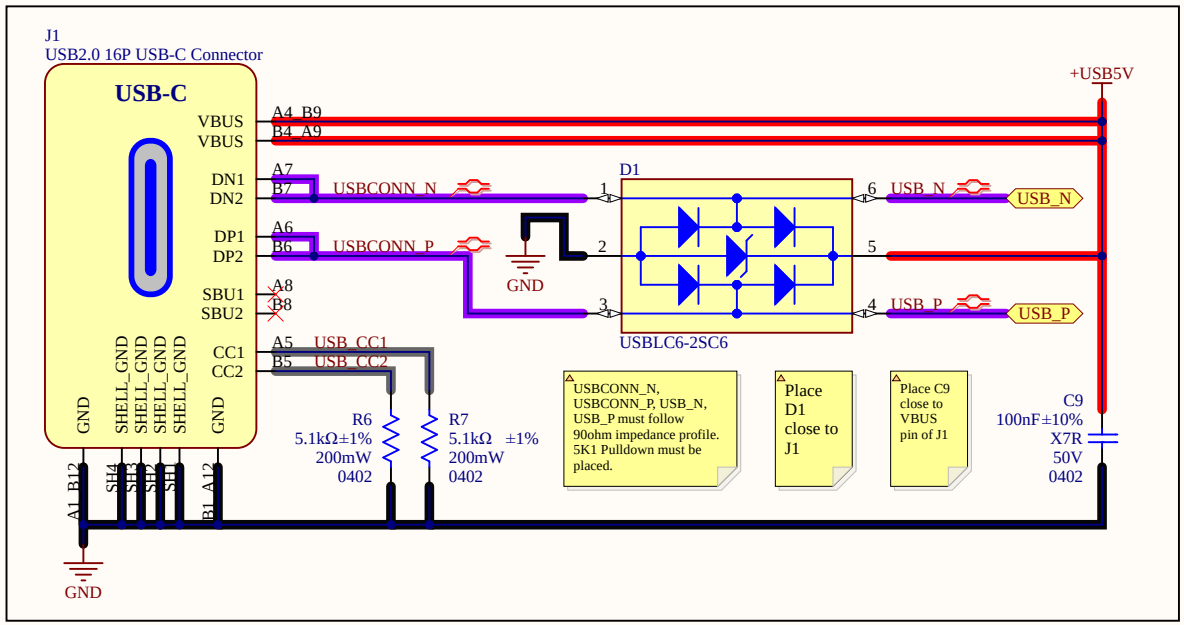
Title: Power		
Size: A4	Number: 3	Revision: 1
Date: 10/07/2026	Sheet 3 of 7	
File: C:\Users\..\Power.SchDoc	Drawn By: Xuyang Liu	

DRV8313RHHR Motor Driver

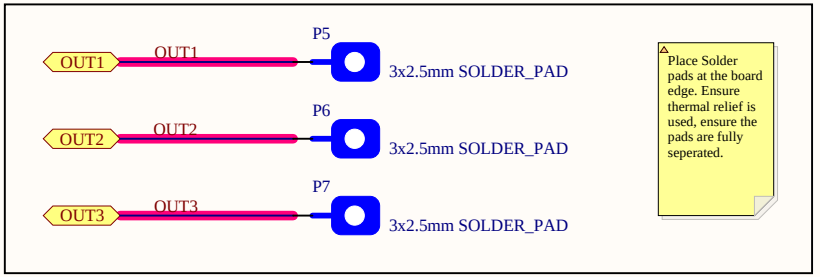


Title: Driver		
Size: A4	Number: 4	Revision: 1
Date: 10/07/2026		Sheet 4 of 7
File: C:\Users\..\Driver.SchDoc		Drawn By: Xuyang Liu

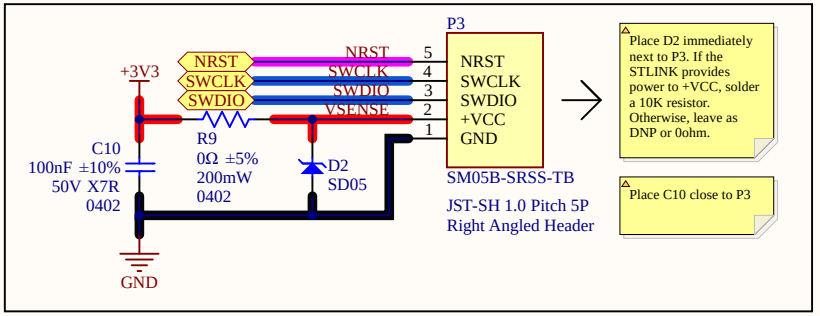
USB-C Data/Power In



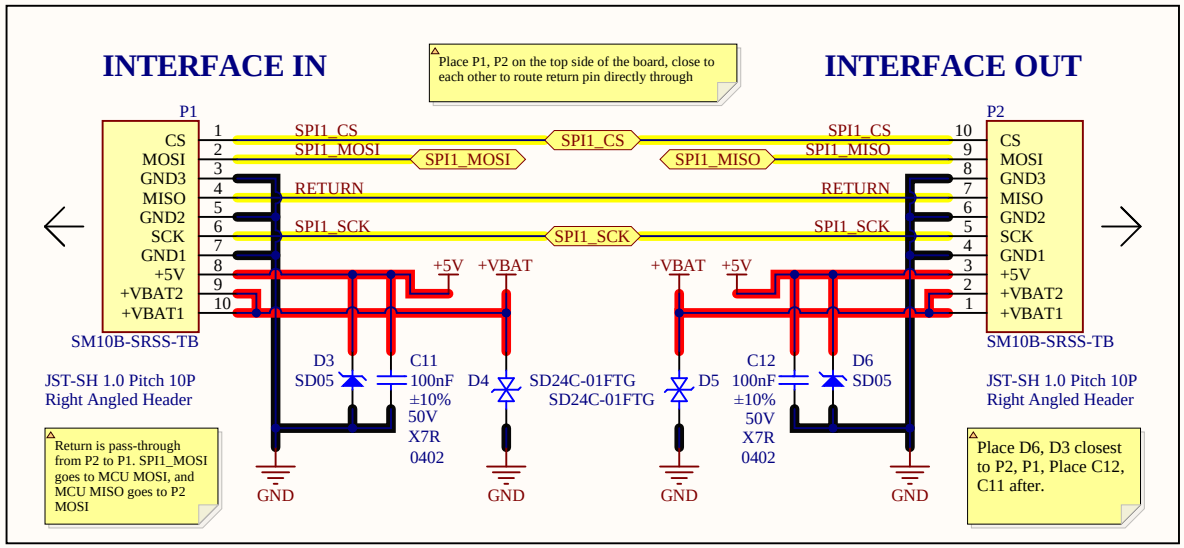
3-Phase Output to BLDC Motor



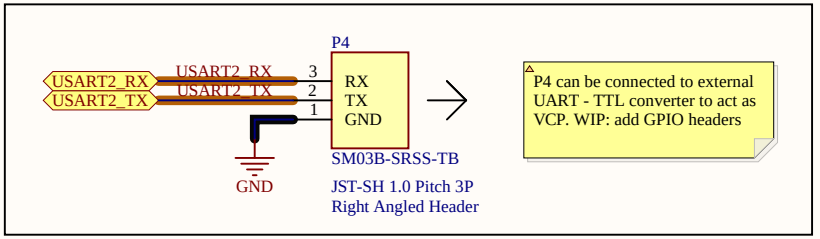
SWD Programming Header



Daisy Chain Power/Data Connector

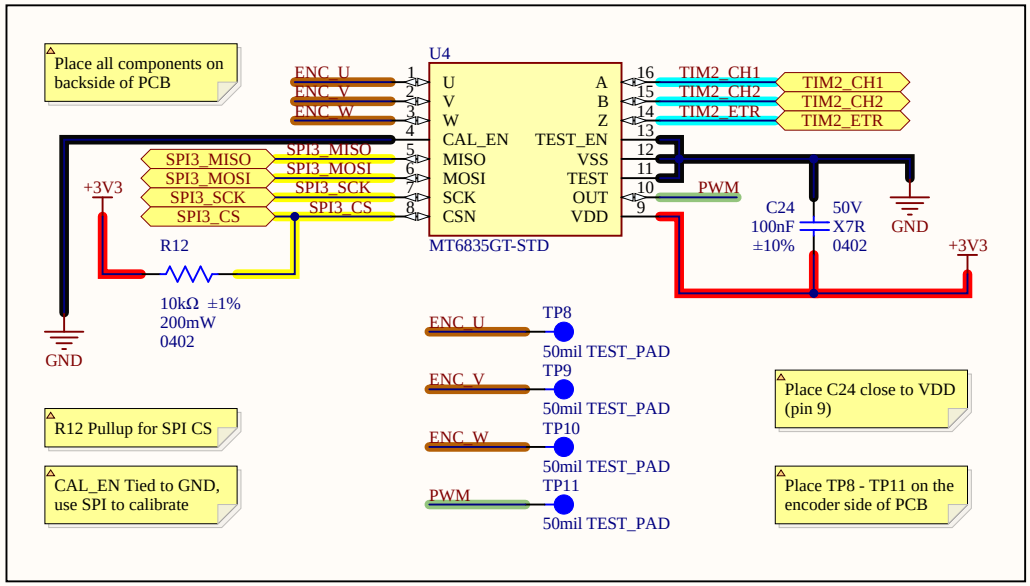


UART Breakout Header



Title: Peripherals		
Size: A4	Number: 5	Revision: 1
Date: 10/07/2026		Sheet 5 of 7
File: C:\Users\..\Peripherals.SchDoc		Drawn By: Xuyang Liu

MT6835 Magnetic Sensor/Encoder



Title: Encoder		
Size: A4	Number: 6	Revision: 1
Date: 10/07/2026		Sheet 6 of 7
File: C:\Users\..\Encoder.SchDoc		Drawn By: Xuyang Liu

M3 Nuts

HW2
Nut, M3.0

HW5
Nut, M3.0

HW8
Nut, M3.0

HW11
Nut, M3.0

M3 Plated Holes

H1
M3 Mounting Holes

H2
M3 Mounting Holes

H3
M3 Mounting Holes

H4
M3 Mounting Holes

M3 Screws

HW3
Screw, M3.0 x 0.6

HW6
Screw, M3.0 x 0.6

HW9
Screw, M3.0 x 0.6

HW12
Screw, M3.0 x 0.6

M3 Spacers

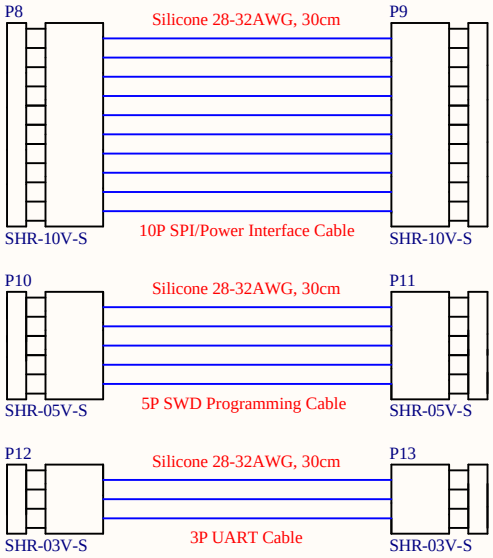
HW1
M3.0 x 05

HW4
M3.0 x 05

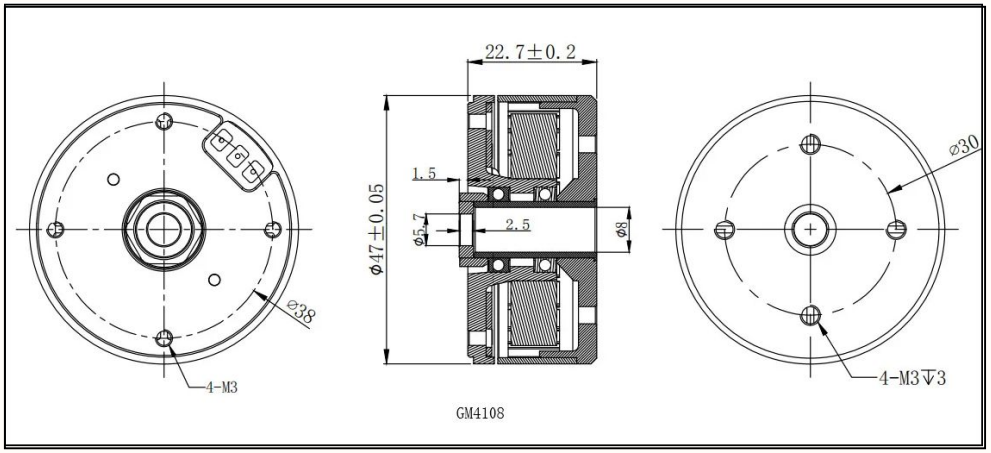
HW7
M3.0 x 05

HW10
M3.0 x 05

JST-SH Connector Housing (3P, 5P, 10P)



GM4108H-120T Gimbal Motor Dimensions

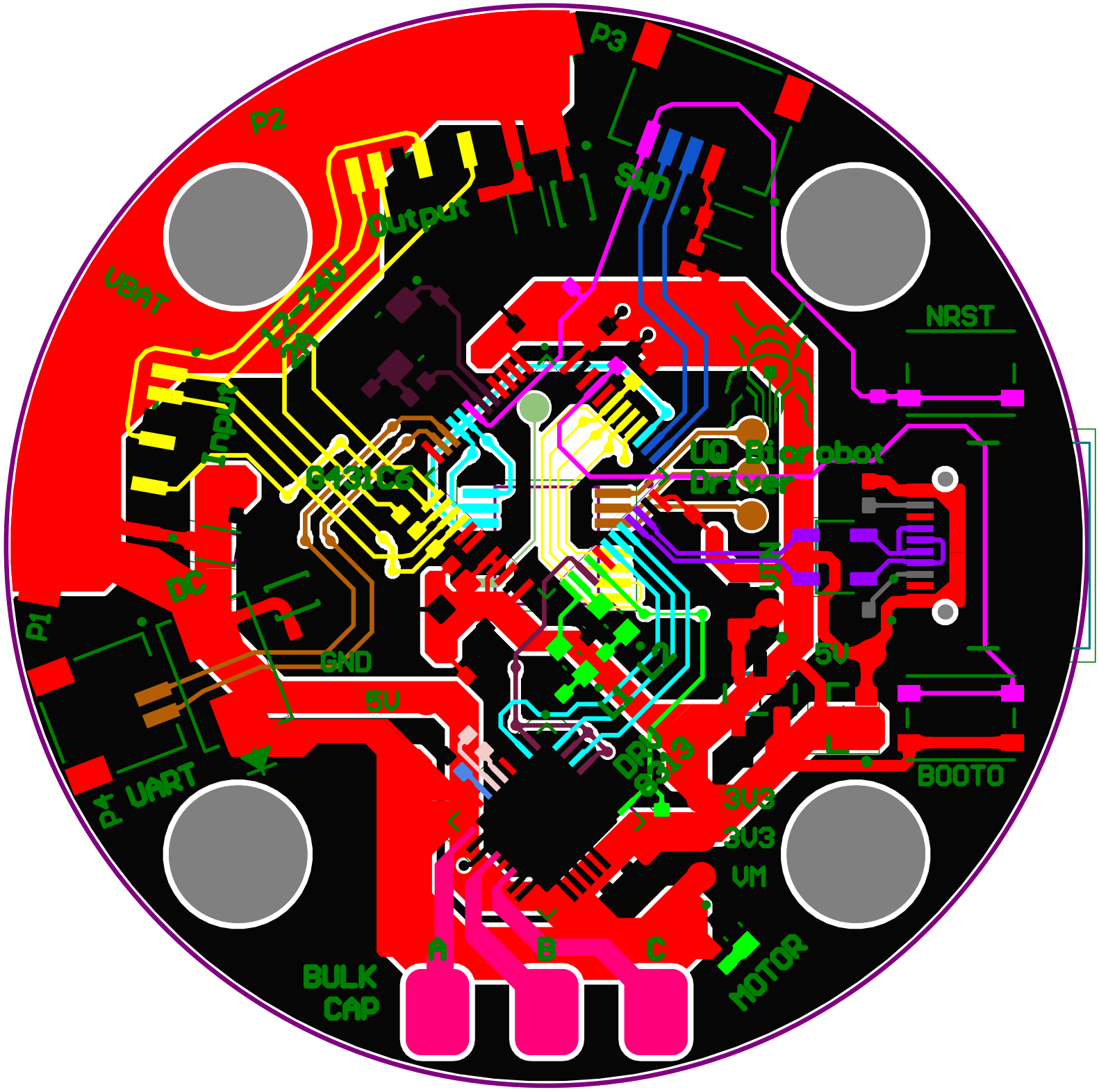


Mounting / Wiring Details

Left: Backside (Stator) of the gimbal motor. Note that 4x M3 mounting holes are required, at a diameter of 38mm. Thus, 4x M3 mounting holes are present on the PCB at +-13.43mm (x,y coordinates). A 10mm OD 5.7mm ID diametrically magnetized ring magnet must be placed in the hexagonal opening to use the onboard MT6835 magnetic sensor. The 3 phase wires can be soldered onto the PCB, the 2.54mm pitch connectors must be stripped.

The JST-SH cable diagram is for one slave board, and therefore only includes the input connector. If the slave board is at the end of the chain, a connector which loops MOSI back to MISO must be used to complete the SPI daisy chain.

Title: Mechanical		
Size: A4	Number: 7	Revision: 1
Date: 10/07/2026		Sheet 7 of 7
File: C:\Users\..\Mechanical.SchDoc		Drawn By: Xuyang Liu


P3
P2
SWD
Output
VBAT
12-24V
Input
NRST
G431C6
UQ Biorobot
Driver
P1
QC
GND
5V
5V
BOOT0
P4
UART
DRV8313
3V3
3V3
VM
A
B
C
BULK
CAP
MOTOR